CS 7545 F26, Homework 2

Zejia Chen
GTID: 904178159

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Exercise 1

(1) Suppose $h^\star = (b_1^\star, \cdots, b_d^\star)$ and let $j_i^\star \in \arg\min_{j\in[R]} \left|\frac{2j-1}{2R} - b_i^\star\right|$. Let $h' = \left(\frac{2j_1^\star-1}{2R}, \cdots, \frac{2j_d^\star-1}{2R}\right)$. Define

$$\ell_i = \min\left(b_i^\star, \frac{2j_i^\star-1}{2R}\right), r_i = \max\left(b_i^\star, \frac{2j_i^\star-1}{2R}\right).$$

Then we have that

$$\begin{aligned}
L(h_R^\star) \le L(h') &= \Pr_{(x,y)}\left[\mathbb{1}\left[\forall i, x_i \le \frac{2j_i^\star-1}{2R}\right] \ne \frac{y+1}{2}\right] \\
&\le \Pr_{(x,y)}\left[\left(\mathbb{1}[\forall i, x_i \le b_i^\star] \ne \frac{y+1}{2}\right) \vee (\exists i, \ell_i < x_i \le r_i)\right] \\
&\le L(h^\star) + \sum_{i=1}^d \Pr_x\left[\ell_i < x_i \le r_i\right] \\
&= L(h^\star) + \sum_{i=1}^d r_i - \ell_i \le L(h^\star) + \frac{d}{2R},
\end{aligned}$$

where the second inequality follows from the union bound and the last inequality follows from $\min_{j\in[R]}\left|\frac{2j-1}{2R} - b_i^\star\right| \le \frac{1}{2R}$ for $i \in [d]$.

(2) By Hoeffding's bound, for any fixed $h \in H_R$, we have that

$$\Pr\left[\hat{L}_S(h) - L(h) \ge \varepsilon\right] \le \exp(-2n\varepsilon^2), \quad \Pr\left[L(h) - \hat{L}_S(h) \ge \varepsilon\right] \le \exp(-2n\varepsilon^2).$$

Therefore, by union bound, with probability at least $1 - 2R^d\exp(-2n\varepsilon^2)$, we have that

$$\forall h \in H_R, L(h) - \hat{L}_S(h) + \hat{L}_S(h_R^\star) - L(h_R^\star) \le 2\varepsilon$$

Since $\hat{L}_S(h_R^\star) \ge \hat{L}_S(\hat{h}_{S,R})$, we have that

$$\begin{aligned}
\Pr\left[L(\hat{h}_{S,R}) - L(h_R^\star) \le 2\varepsilon\right] &\ge \Pr\left[L(\hat{h}_{S,R}) - \hat{L}_S(\hat{h}_{S,R}) + \hat{L}_S(h_R^\star) - L(h_R^\star) \le 2\varepsilon\right] \\
&\ge \Pr\left[\forall h \in H_R, L(h) - \hat{L}_S(h) + \hat{L}_S(h_R^\star) - L(h_R^\star) \le 2\varepsilon\right] \\
&\ge 1 - 2R^d\exp(-2n\varepsilon^2). \qquad (1)
\end{aligned}$$

Plugging $\varepsilon = \sqrt{\frac{d\ln(R)+\ln(2/\delta)}{2n}}$ into Equation (1), we complete the proof.
(3) By Part **(1)** and Part **(2)**, take $R = dn$, with probability at least $1-\delta$, we have that

$$L(\hat{h}_{S,R}) - L(h^\star) \le \frac{d}{2R} + 2\sqrt{\frac{d\ln(R)+\ln(2/\delta)}{2n}} \lesssim \frac{1}{n} + \sqrt{\frac{d\ln(dn)}{n}} = O\left(\sqrt{\frac{d\ln(dn)}{n}}\right).$$

Exercise 2

(1) The VC dimension is 2. For $S \subseteq \mathcal{X}$ with $|S| = 3$, we have that $(1,1,1) \notin H\,|_S$ since the length of the interval in H is only 1.5 and $\sup S - \inf S \ge 2$. For $S = \{1,2\}$, it is clear that $H\,|_S = \{(-1,-1),(-1,1),(1,-1),(1,1)\}$.
(2) $\Pi_H(n) = 2n$. Since the VC dimension is 2, $\Pi_H(n) = 2^n = 2n$ for $n \le 2$. For $n \ge 3$, since the length of the interval in H is only 1.5, it can only cover at most 2 points and these two points must be consecutive. Therefore, for any $S \subset \mathbb{Z}$ with $|S| = n$, we have that

$$|H|_S| = 1 + n + \sum_{x\in S} \mathbb{1}[x+1 \in S] \le 1 + n + n - 1 = 2n,$$

with equality achieved by $S = [n]$. Therefore, $\Pi_H(n) = \max_{|S|=n} |H|_S| = 2n$.
(3) The upper bound of the Sauer-Shelah Lemma is $\frac{n^2+n+2}{2}$, which is quadratic in n, whereas $\Pi_H(n)$ is linear in n. The underlying reason is that the general upper bound misses the combinatorial structure of H. Here if two points in S are labeled $+1$, these two points must be consecutive, which is not generally true for hypothesis class with VC dimension 2. This property restricts $\Pi_H(n)$ to $2n$.

Exercise 3

(1) We have that

$$\begin{aligned}\Pi_{H_1\cup H_2}(n) &= \sup_{|S|=n} |(H_1 \cup H_2)|_S| \le \sup_{|S|=n} (|H_1|_S| + |H_2|_S|) \\ &\le \sup_{|S_1|=n} |H_1|_{S_1}| + \sup_{|S_2|=n} |H_2|_{S_2}| = \Pi_{H_1}(n) + \Pi_{H_2}(n)\end{aligned}$$

(2) By Part **(1)** and the Sauer-Shelah Lemma, we have that for any set S with $|S| = d_1 + d_2 + 2$,

$$\begin{aligned}\Pi_{H_1\cup H_2}(d_1+d_2+2) &\le \Pi_{H_1}(d_1+d_2+2) + \Pi_{H_2}(d_1+d_2+2) \\ &\le \sum_{i=0}^{d_1} \binom{d_1+d_2+2}{i} + \sum_{i=0}^{d_2} \binom{d_1+d_2+2}{(d_1+d_2+2)-i} \\ &= 2^{d_1+d_2+2} - \binom{d_1+d_2+2}{d_1+1} < 2^{d_1+d_2+2}.\end{aligned}$$

Therefore, $d(H_1 \cup H_2) \le d_1 + d_2 + 1$.
(3) For any set $S \subset \mathbb{Z}$ with $|S| = d_1$, we have that $|H_1|_S| = |\{T \subseteq S \mid |T| \le d_1\}| = 2^{d_1}$. However, for any $S \subset \mathbb{Z}$ with $|S| = d_1 + 1$, $(1,1,\cdots,1) \notin H_1|_S$. Therefore, $d(H_1) = d_1$. In the same way, $d(H_2) = d_2$.

Fix $S \subset Z$ with $|S| = d_1 + d_2 + 1$. For any vector v in $\{\pm 1\}^{|S|}$, either v contains at most d_1 entries equal to $+1$ or at most d_2 entries equal to -1. In the former case, there exists some $h \in H_1$ such that $(h(x_1),\cdots,h(x_{|S|})) = v$. In the other case, v can be achieved by some $h \in H_2$. Therefore, $|(H_1 \cup H_2)|_S| = 2^{d_1+d_2+1}$. By Part **(2)**, we conclude that $d(H_1 \cup H_2) = d_1 + d_2 + 1$.

Exercise 4

(1) By the union bound and Hoeffding's bound, we have that

$$\Pr\left[\exists h \in H_k, L(h) - \hat{L}_S(h) \geq \varepsilon\right] \leq \sum_{h \in H_k} \cdot \Pr\left[L(h) - \hat{L}_S(h) \geq \varepsilon\right] \leq \exp(k - 2n\varepsilon^2). \tag{2}$$

Plugging $\varepsilon = \sqrt{\frac{k+\ln(1/\delta)}{2n}}$ into Equation (2), we have that

$$\Pr\left[L(\hat{h}_{S,k}) \leq \hat{L}_S(\hat{h}_{S,k}) + \sqrt{\frac{k+\ln(1/\delta)}{2n}}\right] \geq \Pr\left[\forall h \in H_k, L(h) - \hat{L}_S(h) \leq \sqrt{\frac{k+\ln(1/\delta)}{2n}}\right] \geq 1 - \delta.$$

(2) Since for any hypothesis h, $n\hat{L}_S(h) \in \mathbb{Z}_{\geq 0}$, we have that for $k < k_S$, $\min_{h \in H_{k+1}} \hat{L}_S(h) \leq \min_{h \in H_k} \hat{L}_S(h) - \frac{1}{n}$. The loss function satisfies $\hat{L}_S(h) \in [0, 1]$ for hypothesis h. Therefore, we conclude that

$$0 \leq \min_{h \in H_{k_S}} \hat{L}_S(h) \leq \min_{h \in H_{k_S - 1}} \hat{L}_S(h) - \frac{1}{n} \leq \cdots \leq \min_{h \in H_0} \hat{L}_S(h) - \frac{k_S}{n} \leq 1 - \frac{k_S}{n},$$

which shows $k_S \leq n$.

(3) By union bound and Part **(1)**, for $\delta' > 0$, with probability at least $1 - (n+1)\delta'$,

$$\forall k \in \{0\} \cup [n], \quad L(\hat{h}_{S,k}) \leq \hat{L}_S(\hat{h}_{S,k}) + \sqrt{\frac{k+\ln(1/\delta')}{2n}}.$$

By Part **(2)** and $\hat{h}_{S,\text{iter}} = \hat{h}_{S,k_S}$, and take $\delta = \delta'(n+1)$, we conclude that

$$\begin{aligned}
&\Pr\left[L(\hat{h}_{S,k_S}) \leq \hat{L}_S(\hat{h}_{S,k_S}) + \sqrt{\frac{k_S + \ln(n+1) + \ln(1/\delta)}{2n}}\right] \\
\geq &\Pr\left[\forall k \in \{0\} \cup [n], L(\hat{h}_{S,k}) \leq \hat{L}_S(\hat{h}_{S,k}) + \sqrt{\frac{k + \ln(n+1) + \ln(1/\delta)}{2n}}\right] \geq 1 - \delta.
\end{aligned}$$

Exercise 5

(1) For each point x, sample its label from $\{\pm 1\}$ uniformly at random and the sampling is independent of the algorithm. Let $\mathcal{X} = \{x_1, \cdots, x_d\}$ be the data points and $\mathcal{Y}$ be the corresponding sampled labels. Fix the sample set S, for any output h of $\mathcal{A}(S)$, since $\mathcal{X}$ is shattered by H, we have that for any $x_i \in \mathcal{X}$,

$$x_i \notin S \implies y_i = h(x_i) \text{ w.p. } \frac{1}{2}.$$

Therefore, any excluded point will contribute error $\frac{1}{2}$. From this, we conclude the expectation of the loss function of non-single points:

$$\begin{aligned}
\mathop{\mathbb{E}}_{\mathcal{Y}}\left[\mathop{\mathbb{E}}_{S,\mathcal{A}}\left[L_{\text{non-single}}(\mathcal{A}(S))\right]\right] &\geq \frac{1}{2}(d-1) \cdot \frac{b}{d-1} \cdot \left(1 - \frac{b}{d-1}\right)^n \\
&\geq \frac{1}{2} b \cdot \left(1 - \frac{nb}{d-1}\right),
\end{aligned}$$

where the last inequality follows from the Bernoulli's inequality. Choosing $b = \frac{d-1}{2n}$, we get $\mathbb{E}_{\mathcal{Y}}\left[\mathbb{E}_{S,\mathcal{A}}\left[L_{\text{non-single}}(\mathcal{A}(S))\right]\right] \geq \frac{d-1}{8n}$.

(2) By Part **(1)**, we have that

$$\max_{\mathcal{Y}\in\{\pm 1\}^{\mathcal{X}}} \mathop{\mathbb{E}}_{S,\mathcal{A}}\left[L_{\text{non-single}}(\mathcal{A}(S))\right] \geq \mathop{\mathbb{E}}_{\mathcal{Y}}\left[\mathop{\mathbb{E}}_{S,\mathcal{A}}\left[L_{\text{non-single}}(\mathcal{A}(S))\right]\right] = \frac{b}{2}\left(1 - \frac{nb}{d-1}\right).$$

Fix $\mathcal{Y} \in \arg\max_{\mathcal{Y}\in\{\pm 1\}^{\mathcal{X}}} \mathbb{E}_{S,\mathcal{A}}\left[L_{\text{non-single}}(\mathcal{A}(S))\right]$. We get the distribution $\mathcal{D}$ from Part **(1)**, which is realizable by H. Let $Z = L_{\text{non-single}}(\mathcal{A}(S))$. Since $\mathcal{D}(\text{non-single point}) = b$, we have that $Z \leq b$ and $b - Z$ is nonnegative. Take $b = \frac{d-1}{2n}$, then by Part **(1)** and the choice of $\mathcal{Y}$, $\mathbb{E}[Z] \geq b/4$. By Markov's inequality, we have that

$$\Pr\left[Z \leq \frac{b}{8}\right] = \Pr\left[b - Z \geq \frac{7b}{8}\right] \leq \frac{b - \mathbb{E}[Z]}{7b/8} \leq \frac{6}{7}.$$

Therefore, since $d \geq 2$,

$$\Pr\left[L(\mathcal{A}(S)) < \frac{1}{32}\frac{d}{n}\right] \leq \Pr\left[Z < \frac{1}{32}\frac{d}{n}\right] \leq \Pr\left[Z \leq \frac{1}{16}\frac{d-1}{n}\right] \leq \frac{6}{7}.$$

(3) Let these two points be x_1, x_2 and $\mathcal{D}(x_1) = 1 - b$. Let $S = \{s_1, \cdots, s_n\}$. Since $s_1, \cdots, s_n$ are sampled i.i.d., we have that

$$\Pr\left[\forall i \in [n], s_i = x_1\right] = \prod_{i=1}^{n} \Pr\left[s_i = x_1\right] = \mathcal{D}(x_1)^n = (1-b)^n.$$

As in Part **(1)**, we sample the labels $\mathcal{Y}$ of data points from $\{\pm 1\}$ uniformly at random. If $x_2 \notin S$, then $\mathbb{E}\left[\mathbb{1}[\mathcal{A}(S)(x_2) \neq y_2]\right] = \frac{1}{2}$. Therefore, there exists $y_2' \in \{\pm 1\}$ such that $\Pr_{\mathcal{A}}\left[\mathcal{A}(S)(x_2) \neq y_2'\right] \geq \frac{1}{2}$ (note that here the randomness comes from the algorithm). Hence, we have that

$$\Pr_{S,\mathcal{A}}\left[\mathcal{A}(S)(x_2) \neq y_2'\right] \geq \Pr_{\mathcal{A}}\left[\mathcal{A}(\{x_1, x_1, \cdots, x_1\})(x_2) \neq y_2'\right] \Pr\left[x_2 \notin S\right] \geq (1-b)^n/2.$$

(4) When the other point is incorrectly classified, the loss is at least b ($\mathcal{D}(x_2) = b$). Hence, with probability at least $(1-b)^n/2$, $L(\mathcal{A}(S)) \geq b$.

Take $b = \frac{\ln(1/(2\delta))}{2n}$. Since $\delta \in [e^{-n}, 1/2]$, $b \in [0, \frac{1}{2}]$. Then $(1-b)^n/2 \geq \exp(-2bn)/2$. Therefore,

$$\Pr\left[L(\mathcal{A}(S)) \geq \frac{\ln(1/(2\delta))}{2n}\right] \geq \frac{(1-b)^n}{2} \geq \frac{e^{-2bn}}{2} = \delta.$$